1. The sum of the first n terms of an arithmetic progression is $S_n = 3n^2 + 2n$. What is the 15th term?
A) 85 B) 89 C) 91 D) 95

2. The sum of the first n terms of the series 7 + 12 + 17 + 22 + is 572. Find n.
A) 12 B) 13 C) 14 D) 15

3. Three numbers are in arithmetic progression. Their sum is 21 and their product is 285. What are the numbers?
A) 3, 7, 11 B) 5, 7, 9
C) 4, 7, 10 D) 2, 7, 12

4. If $2 + 5 + 8 + \cdots + (3n - 1) = 440$, find n.
A) 15 B) 16 C) 17 D) 18

5. The sum of the squares of the first n natural numbers is 385. Find n.
A) 8 B) 9 C) 10 D) 11

6. If $1^3 + 2^3 + \ldots + n^3 = 2025$, find n.
A) 8 B) 9 C) 10 D) 11

7. What is the difference between the sum of the cubes and the sum of the squares of the first 10 natural numbers?
A) 2640 B) 2645 C) 2650 D) 2660

8. The sum of the first n terms of the geometric progression 2, 6, 18, ….. is 728. Find n.
A) 5 B) 6 C) 7 D) 8

9. The third term of a geometric progression is 12 and the sixth term is 96. What is the common ratio?
A) 2 B) 3 C) 4 D) 8

10. If $1 + 3 + 9 + \cdots + 3^{n-1} = 364$, find n.
A) 5 B) 6 C) 7 D) 8

11. The first term of a geometric progression is 8 and its common ratio is $\frac{1}{2}$. What is the sum of the first five terms?
A) 15 B) $\frac{31}{2}$ C) $\frac{63}{4}$ D) 16

12. If 3, x, 12 are in geometric progression, what is the positive value of x?
A) 4 B) 5 C) 6 D) 8

13. The first term of a geometric progression is 2 and its common ratio is 3. Which term is equal to 486?
A) 5th B) 6th C) 7th D) 8th

14. If $1 + 2 + 4 + \cdots + 2^{n-1} = 1023$, what is the last term of the series?
A) 256 B) 512 C) 1023 D) 1024

15. If $1^2 + 2^2 + \ldots + n^2 = 385$, what is the value of 1 + 2 + …… + n?
A) 45 B) 50 C) 55 D) 60

Creative Question 1

The 8th term of a geometric progression is $\frac{1}{3}$, and its 9th term is $\frac{1}{81}$. The sum of the first 10 terms of an arithmetic progression is 155, and the sum of its first 20 terms is 610.

(a) Find the value of $1^3 + 2^3 + 3^3 + 4^3 + \ldots\ldots\ldots + 10^3$.
(b) Find the common ratio of the geometric progression.
(c) Find the 35th term of the arithmetic progression.

Creative Question 2

The third term of a geometric progression is $\frac{\sqrt{3}}{4}$, and its sixth term is $\frac{2\sqrt{2}}{27}$. The p-th term of an arithmetic progression is q, and its q-th term is p.

(a) For the series $\frac{1}{\sqrt{2}} + 1 + \sqrt{2} + \ldots\ldots\ldots$, which term is equal to $8\sqrt{2}$?
(b) Find the sum of the first (p + q) terms of the arithmetic progression.
(c) Determine the geometric progression and find the sum of its first five terms.

Creative Question 3

The sum of the first n terms of the series 2+4+8+16+…….is 1022, and the sum of the first 10 terms of another series is 155. The sum of the first 20 terms of that series is 610.

(a) Find the n th term of the series 125+25+5+…….
(b) Determine whether the series 2+4+8+16+…… is a geometric progression and find its common ratio.Using the first series given in the question, find the value of n and then calculate the sum of the squares of the first n natural numbers.
(d) Find the 25th term of the arithmetic progression.

Short Question:

1. Find the common difference of the arithmetic progression: 7, 12, 17, 22,……..
2. Find the 20th term of an arithmetic progression whose first term is 5 and common difference is 3.
3. Find the sum of the first 25 natural numbers.
4. The 5th term of an arithmetic progression is 18 and the 11th term is 42. Find its first term and common difference.
5. Find the common ratio of the geometric progression: 3, 9, 27, 81,……

LESSON BY SHAFIQ